

PAGATEC-INVENTORY-MANAG-BY-TECCEFCORP

Product Vision & Business Requirements Document (Phase 0)

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1. Executive Summary

PAGATEC is a commercial business management platform developed by TECCEF.CORP under the leadership of CEO Yeney B. Herrera. Designed primarily for small and medium-sized businesses (SMBs) in Mexico with expansion intent across Latin America, PAGATEC transcends traditional inventory management to become a complete business operating system.

The platform addresses a critical market gap: existing enterprise solutions (SAP, Oracle NetSuite) are prohibitively complex and expensive for SMBs, while consumer-grade alternatives (Square, QuickBooks, local POS systems) lack the depth, offline reliability, and business intelligence required by growing Latin American businesses. PAGATEC bridges this gap by offering enterprise-grade capability through extreme simplicity, offline-first architecture, and AI-powered decision support.

This document serves as the **constitutional foundation** for all subsequent engineering, design, and business decisions. No feature, module, or architectural choice may be implemented unless it can be traced to a requirement within this document.

2. Project Vision

2.1 Vision Statement

“To become the definitive business operating system for Latin American small and medium businesses — so intuitive that a first-time entrepreneur masters it in 15 minutes, yet so powerful that a multi-store enterprise relies on it for mission-critical operations.”

2.2 Mission Statement

PAGATEC exists to transform how Latin American businesses manage their operations. We do not merely store data; we convert data into actionable intelligence. We do not simply track inventory; we predict shortages before they occur. We do not just record sales; we reveal the patterns that drive profitability.

2.3 Positioning Statement

For small and medium business owners in Latin America who struggle with fragmented tools, unreliable internet connectivity, and lack of business insights, PAGATEC is the all-in-one business management platform that combines offline reliability, cloud synchronization, and AI-powered analytics in a single, affordable solution. Unlike generic international software that ignores local business practices, PAGATEC is built from the ground up for the realities of Latin American commerce — including customer credit (fiado), complex supplier relationships, intermittent connectivity, and local regulatory requirements.

2.4 Vision Principles

- Simplicity is Power:** Every feature must reduce cognitive load, not add it.
- Data is Sacred:** No transaction shall ever be lost, corrupted, or rendered unrecoverable.
- Intelligence is Standard:** AI assistance is not a premium add-on; it is a core expectation.
- Offline is Normal:** The architecture assumes the internet is a luxury, not a guarantee. All core operations function without connectivity.

5. **Local First, Global Ready:** Built for Mexico, architected for Latin America.

3. Strategic Objectives

3.1 Primary Business Objectives

Objective ID	Objective	Target Metric	Timeline
SO-001	Achieve product-market fit in Mexican SMB segment	500 active business subscriptions	Month 12
SO-002	Establish brand recognition as the easiest inventory system to learn	NPS score ≥ 50	Month 6
SO-003	Reduce customer inventory losses by minimum 30%	Customer-reported shrinkage reduction	Month 9
SO-004	Reduce stock shortage incidents by minimum 40%	Automated stock-out alert reduction	Month 9
SO-005	Achieve 99.9% data integrity across all synchronized instances	Zero unrecoverable data loss incidents	Ongoing
SO-006	Expand to 3 additional Latin American countries	Active paying customers in 3 non-Mexico markets	Month 24

3.2 Technical Objectives

Objective ID	Objective	Measurement
TO-001	Sub-2-second load time for all primary screens	Lighthouse / Web Vitals
TO-002	100% offline functionality for core POS and inventory operations	Functional testing without network
TO-003	Conflict-free synchronization across unlimited devices	Automated sync testing
TO-004	AI insight generation within 5 seconds of query	Performance benchmarking
TO-005	99.99% uptime for cloud synchronization services	Uptime monitoring

3.3 Product Objectives

Objective ID	Objective	Measurement
PO-001	New user completes first sale within 15 minutes of account creation	Onboarding funnel analytics
PO-002	User accesses any core report in <= 3 clicks	UX task completion testing
PO-003	Zero training required for basic POS operation	User testing with non-technical subjects
PO-004	AI assistant answers 80% of common business questions correctly	Accuracy benchmarking

4. Target Market & Customer Segments

4.1 Primary Market: Mexico

Mexico represents the initial launch market. Business characteristics:

- **Currency:** Mexican Peso (MXN)
- **Tax Authority:** Servicio de Administracion Tributaria (SAT)
- **Invoicing Requirement:** Comprobante Fiscal Digital por Internet (CFDI) compliance mandatory for formal businesses
- **Connectivity Reality:** Intermittent internet in rural areas; unreliable urban ISP stability
- **Business Culture:** Strong reliance on personal credit (fiado), cash transactions, complex supplier negotiation
- **Language:** Spanish (Mexico variant)

4.2 Secondary Market: Latin America

Planned expansion markets (in priority order):

1. Colombia
2. Chile
3. Peru
4. Argentina
5. Brazil

Each market requires:

- Local tax compliance adaptation
- Local currency support
- Local payment method integration
- Spanish/Portuguese language support

- Regional business practice research

4.3 Customer Segments

Segment A: Micro Retail (1 location, 1-3 employees)

- **Examples:** Corner grocery (tiendita), mini-market, neighborhood hardware store
- **Pain Points:** No formal inventory system, manual bookkeeping, frequent stock-outs, no data insights
- **Decision Maker:** Owner-operator
- **Price Sensitivity:** High
- **Tech Sophistication:** Low

Segment B: Small Retail/Restaurant (1-2 locations, 3-10 employees)

- **Examples:** Small restaurant chain, pharmacy, boutique retail, service business
- **Pain Points:** Using spreadsheets or basic POS, no multi-user permissions, no supplier management
- **Decision Maker:** Owner or general manager
- **Price Sensitivity:** Medium
- **Tech Sophistication:** Medium

Segment C: Growing Multi-Location (3-10 locations, 10-50 employees)

- **Examples:** Regional distributor, multi-store retail, warehouse operations
- **Pain Points:** Disconnected systems per location, no consolidated reporting, complex inventory transfers
- **Decision Maker:** Operations manager or CEO
- **Price Sensitivity:** Low (value-driven)
- **Tech Sophistication:** Medium-High

Segment D: Specialized Businesses

- **Examples:** Pharmacies (regulated inventory), restaurants (perishables), service businesses (appointments + inventory)
- **Pain Points:** Industry-specific compliance, expiration tracking, perishable waste
- **Decision Maker:** Owner or compliance officer
- **Price Sensitivity:** Medium
- **Tech Sophistication:** Medium

5. User Personas

5.1 Persona: Maria Gonzalez — “The First-Time Owner”

- **Role:** Owner, “Abarrotes Maria” (small grocery store)
- **Age:** 42
- **Location:** Guadalajara, Jalisco, Mexico
- **Tech Comfort:** Low. Uses WhatsApp and Facebook. Struggles with Excel.
- **Business:** Single-location family grocery store. 2 employees (husband and cousin).
- **Goals:** Stop running out of milk and eggs. Know which products make money. Stop losing inventory to spoilage.

- **Frustrations:** Current notebook system fails during busy hours. Cannot track who owes money (fiado). Does not understand why some months are profitable and others are not.
- **Success Scenario:** Sets up PAGATEC alone on a Sunday afternoon. By Monday, she scans barcodes instead of writing. By Friday, she sees which products are most profitable. By month-end, she orders inventory based on actual data, not guesswork.
- **Accessibility Needs:** Large touch targets. High contrast mode. Minimal text input.

5.2 Persona: Carlos Mendez — “The Cashier”

- **Role:** Cashier, hardware store
- **Age:** 24
- **Location:** Mexico City
- **Tech Comfort:** Medium. Comfortable with smartphones. Learns apps quickly.
- **Goals:** Process sales fast. Never make pricing mistakes. Handle returns correctly.
- **Frustrations:** Current POS crashes when internet drops. Must memorize prices for unbarcoded items. Cannot check stock without asking the warehouse.
- **Success Scenario:** Opens PAGATEC POS. Scans items. Sees real-time stock levels. Processes cash, card, and fiado payments. Closes shift with exact cash count matching the system.
- **Accessibility Needs:** Keyboard shortcuts. Touch-friendly interface. Works on cheap Android tablet.

5.3 Persona: Ana Reyes — “The Warehouse Manager”

- **Role:** Inventory/Warehouse Manager, multi-store retail
- **Age:** 35
- **Location:** Monterrey, Nuevo Leon
- **Tech Comfort:** Medium-High. Uses inventory apps at previous job.
- **Goals:** Maintain accurate stock counts. Receive purchase orders efficiently. Transfer stock between locations without confusion.
- **Frustrations:** Current system allows negative stock. No barcode support for receiving. Manual transfer sheets get lost.
- **Success Scenario:** Receives purchase order via PAGATEC. Scans each item against expected quantities. System flags discrepancies instantly. Transfers 50 units to Store #2 with one workflow. Inventory is always accurate.
- **Accessibility Needs:** Bulk operations. Barcode scanner integration. Print labels.

5.4 Persona: Roberto Vargas — “The Multi-Store Owner”

- **Role:** CEO, 5-location mini-market chain
- **Age:** 51
- **Location:** Leon, Guanajuato
- **Tech Comfort:** Low-Medium. Uses email and banking apps. Delegates tech decisions.
- **Goals:** See consolidated profit/loss across all stores. Identify top and bottom performing locations. Reduce administrative overhead.
- **Frustrations:** Each store uses a different system. Consolidating data takes 3 days every month. Cannot compare store performance in real-time.
- **Success Scenario:** Opens PAGATEC dashboard. Sees revenue, profit margin, and inventory value for all 5 stores. Drills into Store #3 to see why profit is down. Discovers theft pattern via audit logs. Makes decision in minutes, not days.

- **Accessibility Needs:** Executive dashboard. Automated reports via email. Simple drill-down navigation.

5.5 Persona: Luis Hernandez — “The Restaurant Manager”

- **Role:** Manager, family restaurant
- **Age:** 38
- **Location:** Puebla, Puebla
- **Tech Comfort:** Medium.
- **Goals:** Track ingredient inventory. Calculate dish profitability. Manage supplier deliveries.
- **Frustrations:** Ingredients spoil before use. No visibility into food cost percentage. Suppliers deliver wrong quantities.
- **Success Scenario:** Creates recipes in PAGATEC (1 burger = 150g meat + 1 bun + 30g lettuce). System calculates theoretical vs. actual ingredient usage. Flags 12% variance on meat — investigation reveals portion control issue.
- **Accessibility Needs:** Recipe/bill of materials support. Expiration date tracking. Unit conversion (kg to g, liters to ml).

5.6 Persona: Dra. Elena Castillo — “The Pharmacy Owner”

- **Role:** Owner, community pharmacy
- **Age:** 45
- **Location:** Queretaro, Queretaro
- **Tech Comfort:** Medium.
- **Goals:** Track controlled medications. Ensure compliance. Manage insurance reimbursements.
- **Frustrations:** Manual controlled substance logs. Expired medications discovered too late. No integration with insurance systems.
- **Success Scenario:** PAGATEC tracks batch numbers and expiration dates. Alerts 30 days before expiry. Controlled substance report generated instantly for regulatory inspection.
- **Accessibility Needs:** Batch tracking. Expiration alerts. Regulatory report templates.

6. Core Philosophy & Principles

6.1 Product Philosophy

1. **Simplicity Over Features:** A feature that requires training is a failed feature. Every capability must be discoverable through intuition.
2. **Offline-First:** The architecture assumes the internet is a luxury, not a guarantee. All core operations function without connectivity.
3. **Intelligence by Default:** AI is not a separate module. It permeates every screen, suggesting, warning, and explaining.
4. **Data Immutability:** Business data is a legal record. Nothing is ever truly deleted — it is corrected via audit trail.
5. **Latin American DNA:** Built for fiado, cash-heavy economies, complex supplier relationships, and local regulatory realities.

6.2 Engineering Principles

1. **Reliability Over Speed:** A slow correct answer is better than a fast wrong one.
2. **Security by Design:** Zero-trust architecture. Encrypt everything. Verify everyone.
3. **Progressive Enhancement:** Core features work on a \$100 Android tablet. Advanced features enhance the experience on better hardware.
4. **Testability:** Every requirement must be verifiable. If it cannot be tested, it cannot be shipped.

6.3 Business Principles

1. **Profit Protection:** The software must actively help the business make more money or lose less money. Passive record-keeping is insufficient.
2. **Zero Data Loss:** A lost transaction is a catastrophic failure, not a minor bug.
3. **Traceability:** Every inventory movement, every sale, every financial operation must be traceable to a person, a time, and a reason.
4. **Recoverability:** Disasters happen. The system must be restorable to exact state from backup within 4 hours.
5. **No Vendor Lock-in:** Customer data belongs to the customer. Export must always be possible.

7. Functional Requirements

7.1 Inventory Management Module (INV)

Req ID	Requirement	Priority	Acceptance Criteria
INV-001	System shall support unlimited products per business	Must	Tested with 100,000 product catalog
INV-002	Each product shall have: name, SKU, barcode, category, description, unit of measure, cost price, sale price, tax rate, minimum stock, maximum stock, initial quantity	Must	All fields validated on create/edit
INV-003	System shall support multiple units of measure per product (e.g., box of 24, individual unit) with automatic conversion	Should	Conversion math verified across 1,000 transactions
INV-004	System shall support product variants (size, color, flavor) under a parent product	Should	Parent/child hierarchy functional

Table 5 – continued

Req ID	Requirement	Priority	Acceptance Criteria
INV-005	System shall support barcode scanning via camera and external scanner	Must	Scan-to-add functional on Android tablet
INV-006	System shall track stock levels in real-time across all warehouses/stores	Must	Stock level accurate within 2 seconds of transaction
INV-007	System shall alert when stock falls below minimum threshold	Must	Alert generated within 1 minute of threshold breach
INV-008	System shall support inventory adjustments with mandatory reason code	Must	Every adjustment requires reason; logged in audit trail
INV-009	System shall support physical stock counts with variance reporting	Must	Count sheet -> entry -> variance report workflow functional
INV-010	System shall track product expiration dates and alert 30/15/7 days before expiry	Should	Alert accuracy verified
INV-011	System shall support product images (up to 5 per product)	Should	Image upload, display, delete functional
INV-012	System shall support product status: Active, Inactive, Discontinued	Must	Status changes reflected in POS and reports
INV-013	System shall calculate inventory value by cost method (FIFO, weighted average, LIFO)	Should	Calculation matches manual accounting verification
INV-014	System shall support bulk import via CSV/Excel	Should	1,000 products imported in < 60 seconds
INV-015	System shall support bulk export via CSV/Excel/PDF	Should	Export matches database state exactly
INV-016	System shall track product movement history (in, out, adjustment, transfer)	Must	Complete history retrievable by product
INV-017	System shall support serial number tracking for high-value items	Could	Serial tracked from purchase through sale

Table 5 – continued

Req ID	Requirement	Priority	Acceptance Criteria
INV-018	System shall support batch/lot tracking for regulated industries	Should	Batch traceable from supplier to customer

7.2 Point of Sale Module (POS)

Req ID	Requirement	Priority	Acceptance Criteria
POS-001	System shall provide touch-optimized POS interface	Must	All primary actions accessible via touch on 10" tablet
POS-002	System shall support barcode scanning for rapid item entry	Must	50 items scanned in < 60 seconds
POS-003	System shall support manual product search by name, SKU, or category	Must	Search returns results in < 500ms
POS-004	System shall calculate subtotal, taxes, and total automatically	Must	1,000 test transactions; 100% accuracy
POS-005	System shall support multiple payment methods: Cash, Card, Transfer, Mixed	Must	Each payment type processed and recorded correctly
POS-006	System shall support customer credit (Fiado) sales with credit limit enforcement	Must	Sale blocked if credit limit exceeded
POS-007	System shall generate receipt with business info, items, taxes, payment method, transaction ID	Must	Receipt printable and digital (PDF/email)
POS-008	System shall support receipt reprinting for last transaction	Must	Reprint functional within 24 hours
POS-009	System shall support sale cancellation/void with manager approval and reason	Must	Void requires supervisor PIN; logged
POS-010	System shall support returns and exchanges with original transaction reference	Must	Return links to original sale; inventory restored

Table 6 – continued

Req ID	Requirement	Priority	Acceptance Criteria
POS-011	System shall support discounts (percentage or fixed) per item or per ticket	Should	Discount authorized per role settings
POS-012	System shall support promotions (buy X get Y, bundle pricing)	Could	Promotion engine functional
POS-013	System shall operate in offline mode with local queue	Must	100 transactions processed offline; synced correctly
POS-014	System shall display real-time stock level during sale	Should	Stock visible without leaving POS screen
POS-015	System shall support shift management (open/close with cash count)	Must	Shift reconciliation balanced or flagged
POS-016	System shall support multiple POS stations per store	Must	5 stations operational simultaneously
POS-017	System shall support quick-keys for top-selling items	Should	Customizable quick-key grid
POS-018	System shall generate end-of-shift report (sales, returns, voids, cash count)	Must	Report accuracy matches transaction log

7.3 Purchasing Module (PUR)

Req ID	Requirement	Priority	Acceptance Criteria
PUR-001	System shall support purchase order creation with supplier, items, quantities, expected delivery date	Must	PO created, editable, printable
PUR-002	System shall track PO status: Draft, Sent, Partial, Received, Cancelled	Must	Status transitions logged
PUR-003	System shall support goods receipt against PO with variance capture	Must	Over/shortage flagged at receipt

Table 7 – continued

Req ID	Requirement	Priority	Acceptance Criteria
PUR-004	System shall update inventory and cost of goods upon receipt	Must	Inventory and COGS accurate post-receipt
PUR-005	System shall support direct purchases (without PO)	Should	Direct purchase workflow functional
PUR-006	System shall track supplier invoices and payment status	Should	Invoice linked to PO; payment status visible
PUR-007	System shall generate purchase history and supplier performance reports	Should	Report shows on-time delivery, price variance

7.4 Supplier Management Module (SUP)

Req ID	Requirement	Priority	Acceptance Criteria
SUP-001	System shall maintain supplier master data: name, contact, address, tax ID, payment terms, lead time	Must	All fields validated
SUP-002	System shall link suppliers to products they provide	Should	Product-supplier association functional
SUP-003	System shall track supplier purchase history and payment history	Should	History accurate across all transactions
SUP-004	System shall support supplier rating/notes	Could	Rating stored and visible

7.5 Customer Management Module (CUS)

Req ID	Requirement	Priority	Acceptance Criteria
CUS-001	System shall maintain customer master data: name, phone, email, address, tax ID (RFC)	Must	RFC format validated for Mexico
CUS-002	System shall track customer purchase history	Must	History complete and searchable

Table 9 – continued

Req ID	Requirement	Priority	Acceptance Criteria
CUS-003	System shall support customer segmentation (VIP, regular, new)	Could	Segmentation functional
CUS-004	System shall support customer loyalty points program	Could	Points accrue and redeem correctly

7.6 Fiado (Customer Credit) Module (FIA)

Req ID	Requirement	Priority	Acceptance Criteria
FIA-001	System shall allow credit sales with per-customer credit limit	Must	Limit enforced at POS
FIA-002	System shall track outstanding balance per customer	Must	Balance accurate to the centavo
FIA-003	System shall support partial payments against credit balance	Must	Payment reduces balance; history kept
FIA-004	System shall generate customer statement (purchases, payments, balance)	Must	Statement printable and digital
FIA-005	System shall alert when customer approaches or exceeds credit limit	Should	Alert at 80% and 100% of limit
FIA-006	System shall support credit terms (due dates, interest)	Could	Terms configurable per customer
FIA-007	System shall generate aging report (current, 30, 60, 90+ days overdue)	Should	Aging accurate per customer

7.7 Financial Dashboard Module (FIN)

Req ID	Requirement	Priority	Acceptance Criteria
FIN-001	Dashboard shall display daily sales, expenses, profit	Must	Values update in real-time
FIN-002	Dashboard shall display inventory value (cost basis)	Must	Calculation matches inventory module

Table 11 – continued

Req ID	Requirement	Priority	Acceptance Criteria
FIN-003	Dashboard shall display cash position per store	Must	Cash on hand accurate
FIN-004	Dashboard shall display top-selling products today/this week/this month	Should	Ranking accurate
FIN-005	Dashboard shall display low-stock alerts	Must	Alerts match inventory thresholds
FIN-006	Dashboard shall be customizable per user role	Should	Owner sees everything; cashier sees sales only

7.8 Reports Module (RPT)

Req ID	Requirement	Priority	Acceptance Criteria
RPT-001	System shall generate daily sales report (items, quantities, revenue, payment methods)	Must	Report matches transaction data
RPT-002	System shall generate weekly sales summary	Must	Aggregated correctly by week
RPT-003	System shall generate monthly profit/loss statement	Must	Revenue - COGS - Expenses = Profit
RPT-004	System shall generate inventory valuation report	Must	Valuation method applied consistently
RPT-005	System shall generate low-stock report	Must	All items below threshold included
RPT-006	System shall generate best/worst seller report	Should	Ranking by quantity and revenue
RPT-007	System shall generate supplier purchase report	Should	PO and direct purchases included
RPT-008	System shall generate customer credit (Fiado) report	Must	All credit customers with balances
RPT-009	System shall generate tax report (IVA collected, IVA paid)	Must	Tax calculations accurate for SAT
RPT-010	System shall generate employee performance report (sales per cashier)	Should	Shift data aggregated by employee

Table 12 – continued

Req ID	Requirement	Priority	Acceptance Criteria
RPT-011	All reports shall be exportable to PDF and Excel	Should	Export formatting correct
RPT-012	All reports shall support date range filtering	Must	Filter applies correctly to data

7.9 Analytics Module (ANL)

Req ID	Requirement	Priority	Acceptance Criteria
ANL-001	System shall display sales trends (line chart, daily/weekly/monthly)	Should	Chart renders accurately
ANL-002	System shall display inventory turnover rate	Should	Calculation: COGS / Average Inventory
ANL-003	System shall display profit margin by product/category	Should	Margin = (Revenue - COGS) / Revenue
ANL-004	System shall display seasonal demand patterns	Could	Historical comparison by month
ANL-005	System shall support custom date range analytics	Should	Range applies to all charts

7.10 Employee Management Module (EMP)

Req ID	Requirement	Priority	Acceptance Criteria
EMP-001	System shall maintain employee records: name, role, contact, hire date, status	Must	CRUD operations functional
EMP-002	System shall support multiple roles per employee	Should	Role assignment functional
EMP-003	System shall track employee sales performance	Should	Performance linked to shift/POS transactions
EMP-004	System shall support employee schedule/attendance (basic)	Could	Clock-in/clock-out recorded

7.11 Role & Permission System Module (SEC)

Req ID	Requirement	Priority	Acceptance Criteria
SEC-001	System shall support role-based access control (RBAC)	Must	Roles: Owner, Manager, Cashier, Warehouse, Accountant, Read-Only
SEC-002	System shall allow custom permission granularity per role	Should	Per-module, per-action permissions
SEC-003	System shall enforce permissions at API and UI level	Must	API rejects unauthorized requests; UI hides unauthorized actions
SEC-004	System shall support audit logging of all user actions	Must	Log: user, action, timestamp, IP, before/after values
SEC-005	System shall require strong passwords (min 8 chars, complexity)	Must	Password policy enforced
SEC-006	System shall support session timeout after inactivity	Should	Timeout configurable

7.12 Multi-Warehouse Module (WHS)

Req ID	Requirement	Priority	Acceptance Criteria
WHS-001	System shall support multiple warehouses per business	Must	Warehouse creation and management functional
WHS-002	System shall track stock per warehouse	Must	Stock levels warehouse-specific
WHS-003	System shall support stock transfers between warehouses with approval	Must	Transfer reduces source, increases destination; logged
WHS-004	System shall support warehouse-specific reorder points	Could	Thresholds per warehouse

7.13 Multi-Store Module (MST)

Req ID	Requirement	Priority	Acceptance Criteria
MST-001	System shall support multiple stores per business account	Must	Store creation and management functional
MST-002	System shall support consolidated reporting across stores	Must	Report aggregates all stores

Table 17 – continued

Req ID	Requirement	Priority	Acceptance Criteria
MST-003	System shall support store-specific inventory, employees, and settings	Must	Data isolation per store
MST-004	System shall support inter-store transfers	Should	Transfer workflow functional
MST-005	System shall support store-level permissions (manager sees only their store)	Should	Data access restricted by store assignment

7.14 AI Business Assistant Module (AIA)

Req ID	Requirement	Priority	Acceptance Criteria
AIA-001	AI shall answer natural language questions about business data	Should	80% accuracy on common questions
AIA-002	AI shall generate sales predictions based on historical data	Could	Prediction within 20% of actual
AIA-003	AI shall recommend reorder quantities and timing	Could	Recommendation reduces stock-outs by 20%
AIA-004	AI shall identify slow-moving and dead stock	Should	Identification matches manual analysis
AIA-005	AI shall provide profit analysis insights (top margins, loss leaders)	Should	Analysis accurate per product
AIA-006	AI shall generate plain-language business summaries	Could	Summary generated weekly
AIA-007	AI responses shall reference actual data points, not hallucinate	Must	Every claim traceable to database record

7.15 Notifications Module (NTF)

Req ID	Requirement	Priority	Acceptance Criteria
NTF-001	System shall send low-stock alerts	Must	Alert delivered within 1 minute of threshold breach
NTF-002	System shall send expiration alerts	Should	Alert 30/15/7 days before expiry

Table 19 – continued

Req ID	Requirement	Priority	Acceptance Criteria
NTF-003	System shall send daily summary email to owner	Could	Email sent at configurable time
NTF-004	System shall send sync conflict alerts	Should	Alert when manual resolution required
NTF-005	Notifications shall support in-app, email, and WhatsApp (future)	Should	In-app mandatory; others configurable

7.16 Settings Module (SET)

Req ID	Requirement	Priority	Acceptance Criteria
SET-001	System shall allow business profile configuration (name, address, tax info, logo)	Must	Profile saved and displayed on receipts
SET-002	System shall allow tax configuration (IVA rate, exempt categories)	Must	Tax calculation matches configuration
SET-003	System shall allow currency and number format configuration	Must	MXN format default; configurable
SET-004	System shall allow receipt template customization	Should	Header, footer, message customizable
SET-005	System shall support backup configuration (frequency, retention)	Should	Backup runs per schedule
SET-006	System shall support dark mode / light mode toggle	Should	Theme switch functional
SET-007	System shall support language switch (Spanish / English)	Should	All UI elements translated

7.17 Offline Synchronization Module (OFF)

Req ID	Requirement	Priority	Acceptance Criteria
OFF-001	System shall operate fully without internet connectivity	Must	All core features functional offline
OFF-002	System shall queue transactions locally during offline periods	Must	Queue stores all transactions without loss

Table 21 – continued

Req ID	Requirement	Priority	Acceptance Criteria
OFF-003	System shall synchronize queued transactions when connectivity returns	Must	Sync completes successfully; no duplicates
OFF-004	System shall detect and flag sync conflicts	Must	Conflict identified with clear resolution options
OFF-005	System shall resolve simple conflicts automatically (last-write-wins with timestamp)	Should	Auto-resolution for non-critical conflicts
OFF-006	System shall require manual resolution for complex conflicts (inventory, pricing)	Must	User prompted to choose correct value
OFF-007	System shall display sync status (online/offline/pending)	Must	Status visible and accurate
OFF-008	System shall support periodic background sync when online	Should	Sync interval configurable

7.18 Cloud Synchronization Module (CLD)

Req ID	Requirement	Priority	Acceptance Criteria
CLD-001	System shall synchronize data across all authorized devices in real-time	Must	Data consistent across devices within 5 seconds
CLD-002	System shall encrypt all data in transit (TLS 1.3)	Must	Security audit passed
CLD-003	System shall encrypt all data at rest (AES-256)	Must	Encryption verified
CLD-004	System shall support automatic cloud backup	Must	Daily backup minimum
CLD-005	System shall support data restore from backup	Must	Restore completes within 4 hours
CLD-006	System shall maintain data integrity during sync (checksums, transaction logs)	Must	Zero corruption incidents in testing

7.19 Audit Logs Module (ADT)

Req ID	Requirement	Priority	Acceptance Criteria
ADT-001	System shall log every create, update, delete operation	Must	Log entry for every mutation
ADT-002	Audit log shall include: timestamp, user, action, entity, old value, new value, IP address	Must	All fields populated
ADT-003	Audit logs shall be immutable and non-deletable by users	Must	Even admins cannot delete logs
ADT-004	Audit logs shall be searchable and filterable	Should	Search by user, date, entity type
ADT-005	Audit logs shall be exportable	Could	CSV/PDF export functional

7.20 Backups Module (BKP)

Req ID	Requirement	Priority	Acceptance Criteria
BKP-001	System shall perform automatic daily cloud backup	Must	Backup completes without error
BKP-002	System shall retain backups for minimum 30 days	Should	Retention policy enforced
BKP-003	System shall allow manual backup trigger	Should	Manual backup functional
BKP-004	System shall allow backup restore to specific point in time	Should	Restore accurate to selected backup

7.21 Import/Export Module (IEX)

Req ID	Requirement	Priority	Acceptance Criteria
IEX-001	System shall support bulk product import via CSV/Excel	Should	Template provided; validation errors reported
IEX-002	System shall support bulk product export via CSV/Excel/PDF	Should	Export matches current data
IEX-003	System shall support customer/supplier import/export	Could	Same format as products

8. Non-Functional Requirements

8.1 Performance Requirements

Req ID	Requirement	Target
NFR-PER-001	Initial application load time	< 3 seconds on 4G connection
NFR-PER-002	Screen-to-screen navigation	< 1 second
NFR-PER-003	Search response time	< 500ms for 10,000 products
NFR-PER-004	Report generation (daily)	< 5 seconds
NFR-PER-005	Report generation (monthly)	< 30 seconds
NFR-PER-006	POS transaction processing	< 2 seconds per item
NFR-PER-007	Sync operation (100 transactions)	< 10 seconds
NFR-PER-008	Concurrent users per store	Minimum 20 simultaneous
NFR-PER-009	Database query response (p95)	< 200ms

8.2 Availability & Reliability Requirements

Req ID	Requirement	Target
NFR-AVL-001	Cloud sync service uptime	99.9% (max 8.76 hours downtime/year)
NFR-AVL-002	Offline functionality	100% of core features
NFR-AVL-003	Data loss tolerance	Zero unrecoverable transactions
NFR-AVL-004	Mean Time Between Failures (MTBF)	> 720 hours
NFR-AVL-005	Mean Time To Recovery (MTTR)	< 4 hours

8.3 Security Requirements

Req ID	Requirement	Standard
NFR-SEC-001	Password storage	bcrypt/Argon2 hashing
NFR-SEC-002	Data in transit encryption	TLS 1.3 minimum
NFR-SEC-003	Data at rest encryption	AES-256
NFR-SEC-004	API authentication	JWT with refresh token rotation
NFR-SEC-005	Session management	Secure httpOnly cookies; 30-min timeout
NFR-SEC-006	Input validation	OWASP Top 10 mitigation
NFR-SEC-007	SQL injection prevention	Parameterized queries exclusively
NFR-SEC-008	XSS prevention	Content Security Policy; output encoding
NFR-SEC-009	Rate limiting	100 requests/minute per IP; 1000 per user
NFR-SEC-010	Role-based access enforcement	Every endpoint validates permissions

8.4 Scalability Requirements

Req ID	Requirement	Target
NFR-SCL-001	Products per business	100,000+
NFR-SCL-002	Transactions per day per store	10,000+
NFR-SCL-003	Concurrent stores per business	50+
NFR-SCL-004	Users per business	200+
NFR-SCL-005	Historical data retention	7 years minimum

8.5 Usability Requirements

Req ID	Requirement	Target
NFR-USA-001	New user time-to-first-sale	< 15 minutes
NFR-USA-002	Task completion rate (core workflows)	> 95% without documentation
NFR-USA-003	Interface language	Spanish (Mexico) primary; English secondary
NFR-USA-004	Touch target size	Minimum 44x44px
NFR-USA-005	Color contrast	WCAG AA compliance
NFR-USA-006	Screen reader support	WCAG 2.1 Level AA
NFR-USA-007	Keyboard navigation	All features accessible without mouse

8.6 Maintainability Requirements

Req ID	Requirement	Target
NFR-MNT-001	Code test coverage	> 80%
NFR-MNT-002	API documentation	OpenAPI/Swagger; 100% endpoint coverage
NFR-MNT-003	Deployment time	< 30 minutes
NFR-MNT-004	Feature toggle system	All major features toggleable

8.7 Compliance Requirements

Req ID	Requirement	Jurisdiction
NFR-CMP-001	CFDI (Comprobante Fiscal Digital por Internet) compliance	Mexico
NFR-CMP-002	SAT (Servicio de Administracion Tributaria) reporting	Mexico
NFR-CMP-003	IVA (Impuesto al Valor Agregado) calculation accuracy	Mexico
NFR-CMP-004	Data privacy (LFPDPPP)	Mexico
NFR-CMP-005	Controlled substance tracking (pharmacies)	Mexico —COFEPRIS

Table 32 – continued

Req ID	Requirement	Jurisdiction
NFR-CMP-006	LGPD readiness	Brazil (future)
NFR-CMP-007	GDPR readiness	EU customers (future)

9. Business Workflows

9.1 Product Onboarding Workflow

1. **Actor:** Owner/Manager
2. **Trigger:** New product needs to be sold
3. **Steps:**
 - Navigate to Inventory -> Products -> New
 - Enter product name, select category
 - Enter or scan barcode (auto-lookup if barcode exists in database)
 - Set unit of measure, cost price, sale price, tax rate
 - Set minimum and maximum stock levels
 - Enter initial quantity and warehouse location
 - Upload product image (optional)
 - Save; system generates SKU if not provided
4. **Post-conditions:** Product available in POS and inventory reports
5. **Exception:** Duplicate barcode detected -> prompt to merge or differentiate

9.2 Purchase Order Workflow

1. **Actor:** Manager/Warehouse
2. **Trigger:** Inventory below reorder point or scheduled restock
3. **Steps:**
 - System generates low-stock alert or user creates manual PO
 - Select supplier from master list
 - Add products; system suggests quantities based on sales velocity
 - Review and adjust quantities, prices, delivery date
 - Submit PO; status = Draft
 - Send to supplier (print/email/export)
 - Upon delivery, open PO -> Receive Goods
 - Scan/enter received quantities; system flags variances
 - Confirm receipt; inventory updated; cost of goods recalculated
 - Supplier invoice recorded and linked to PO
4. **Post-conditions:** Stock increased; PO status = Received; payable recorded
5. **Exception:** Partial delivery -> PO status = Partial; remaining items tracked

9.3 Sales / POS Workflow

1. **Actor:** Cashier
2. **Trigger:** Customer presents items for purchase
3. **Steps:**

- Cashier opens POS screen
 - Scans or searches for each item
 - System displays item name, price, running total
 - Apply discounts if authorized and applicable
 - Select payment method: Cash, Card, Transfer, Mixed, or Fiado
 - If Fiado: select customer; system validates credit limit
 - If Mixed: enter amounts per method
 - System calculates change (if cash)
 - Confirm sale; receipt prints/emails
 - Inventory decremented in real-time
4. **Post-conditions:** Sale recorded; inventory reduced; payment recorded; receipt generated
 5. **Exception:** Item out of stock -> warning displayed; cashier decides to proceed or cancel

9.4 Inventory Adjustment Workflow

1. **Actor:** Warehouse Manager
2. **Trigger:** Physical count differs from system; damage; theft; expiration
3. **Steps:**
 - Navigate to Inventory -> Adjustments -> New
 - Select product and warehouse
 - Enter actual quantity vs. system quantity
 - Select reason: Physical Count, Damage, Theft, Expiration, Other
 - Enter detailed notes
 - Submit; system requires manager-level approval if variance > threshold
 - Upon approval, inventory updated; audit log records old/new values
4. **Post-conditions:** Inventory corrected; reason documented; audit trail complete
5. **Exception:** Negative stock would result -> system blocks or warns per configuration

9.5 Customer Credit (Fiado) Workflow

1. **Actor:** Cashier + Customer
2. **Trigger:** Customer requests credit purchase
3. **Steps:**
 - At POS, cashier selects “Fiado” payment method
 - Search/select customer from database
 - System displays current balance and credit limit
 - If sale amount + current balance <= credit limit: proceed
 - If exceeds limit: system blocks; cashier informs customer
 - Complete sale; balance updated
 - Customer signs digital receipt or paper receipt
4. **Post-conditions:** Sale recorded as credit; customer balance increased
5. **Exception:** New customer without credit account -> prompt to create with owner approval

9.6 Credit Payment (Fiado Collection) Workflow

1. **Actor:** Cashier/Manager
2. **Trigger:** Customer makes payment against credit balance
3. **Steps:**
 - Navigate to Customers -> [Customer] -> Make Payment

- Display current balance and outstanding invoices
 - Enter payment amount and method (cash, transfer, card)
 - Apply payment to specific invoices or general balance
 - Record payment; receipt generated
 - Balance updated; aging recalculated
4. **Post-conditions:** Payment recorded; balance reduced; receipt issued

9.7 Multi-Store Transfer Workflow

1. **Actor:** Warehouse Manager / Store Manager
2. **Trigger:** Store A has excess stock; Store B has shortage
3. **Steps:**
 - Navigate to Inventory -> Transfers -> New
 - Select source warehouse/store and destination
 - Add products and quantities
 - System validates source has sufficient stock
 - Submit transfer request; status = Pending
 - Destination manager approves/rejects
 - If approved: source stock reduced; destination stock increased
 - Transfer logged; audit trail complete
4. **Post-conditions:** Stock rebalanced; both locations updated
5. **Exception:** Source stock insufficient at time of approval -> transfer blocked; notification sent

9.8 Offline Sync Recovery Workflow

1. **Actor:** System (automatic) + User (if conflict)
2. **Trigger:** Device reconnects to internet after offline period
3. **Steps:**
 - System detects connectivity restoration
 - Uploads local transaction queue to cloud
 - Downloads updates from other devices
 - Compares timestamps and versions
 - Simple conflicts (different fields): auto-resolve via last-write-wins
 - Complex conflicts (same inventory item, same field): flag for user
 - User receives notification: "Sync conflict requires resolution"
 - User reviews conflict details (local vs. server values)
 - User selects correct value; system applies and logs resolution
 - Sync status returns to "Synchronized"
4. **Post-conditions:** All devices consistent; no data loss
5. **Exception:** Sync interrupted -> resume from checkpoint; no duplicate uploads

9.9 End-of-Shift Workflow

1. **Actor:** Cashier
2. **Trigger:** Shift end (scheduled or manual)
3. **Steps:**
 - Cashier selects "Close Shift"
 - System displays: shift start time, total sales, total returns, total voids, expected cash
 - Cashier counts physical cash in drawer

- Enters counted amount
 - System calculates variance (expected vs. actual)
 - If variance within tolerance: shift closed; report generated
 - If variance outside tolerance: manager approval required
 - Shift report saved; new shift can be opened
4. **Post-conditions:** Shift closed; cash accountability recorded

9.10 AI Insight Generation Workflow

1. **Actor:** Business Owner / Manager
2. **Trigger:** User asks question or scheduled insight generation
3. **Steps:**
 - User asks natural language question (e.g., “What should I reorder this week?”)
 - AI parses intent and identifies relevant data entities
 - AI queries database for historical sales, current stock, lead times
 - AI applies prediction model or analytical logic
 - AI generates response with specific data references
 - Response displayed with confidence level and data sources
 - User can drill down into supporting data or refine question
4. **Post-conditions:** Insight delivered; user makes informed decision
5. **Exception:** Insufficient data -> AI responds with “I need more data to answer accurately” and specifies what is missing

10. Data Requirements (High-Level)

10.1 Data Entities (Conceptual)

The system shall manage the following conceptual data domains:

1. **Business Entity:** Company profile, tax information, branding, subscription status
2. **User Entity:** Employees, roles, permissions, authentication credentials, session data
3. **Product Entity:** Catalog, variants, barcodes, pricing, costs, taxes, images, units of measure
4. **Inventory Entity:** Stock levels, warehouse locations, movements, adjustments, transfers, counts
5. **Transaction Entity:** Sales, purchases, returns, payments, credit transactions
6. **Customer Entity:** Profiles, credit limits, balances, purchase history, loyalty data
7. **Supplier Entity:** Profiles, payment terms, purchase history, performance metrics
8. **Financial Entity:** Accounts, tax records, shift reconciliations, expense categories
9. **Audit Entity:** Action logs, change history, system events, security incidents
10. **AI Entity:** Queries, responses, data references, feedback, model configurations

10.2 Data Volume Estimates (Per Business, Year 1)

Entity	Estimated Volume	Growth Rate
Products	1,000 - 50,000	Linear with business growth
Transactions	50,000 - 500,000	Daily operations
Customers	500 - 20,000	Depends on loyalty program adoption

Table 33 – continued

Entity	Estimated Volume	Growth Rate
Audit Logs	500,000 - 5,000,000	High; every action logged
AI Interactions	1,000 - 10,000	Depends on feature adoption

10.3 Data Retention Policies

Data Type	Retention Period	Rationale
Transaction data	7 years	Tax compliance (SAT requirement)
Audit logs	7 years	Legal traceability
Customer data	Duration of relationship + 2 years	LFPDPPP compliance
Backup data	30 days minimum	Disaster recovery
Session logs	90 days	Security analysis
AI query logs	1 year	Model improvement and debugging

10.4 Data Quality Requirements

1. **Accuracy:** Financial calculations must be accurate to 2 decimal places (MXN centavos)
2. **Consistency:** Inventory quantities must never be negative unless explicitly configured to allow it
3. **Completeness:** Every transaction must have: timestamp, user, items, amounts, payment method
4. **Timeliness:** Sync latency must not exceed 5 seconds for online operations
5. **Validity:** Barcodes must conform to EAN-13 or UPC-A standards; RFC must conform to SAT format

11. Integration Requirements

11.1 Internal Integrations

Integration	Description	Priority
POS <-> Inventory	Real-time stock deduction on sale	Must
POS <-> Customer Credit	Balance validation and update	Must
Inventory <-> Purchasing	Reorder point triggers PO suggestions	Should
AI <-> All Modules	Data access for insights and predictions	Should
Reports <-> All Modules	Aggregation across all data sources	Must

11.2 External Integrations (Current Phase)

Integration	Description	Priority
SAT/CFDI	Electronic invoicing compliance	Must (Mexico)

Table 36 – continued

Integration	Description	Priority
Email Service	Transaction receipts, reports, alerts	Should
SMS/WhatsApp	Notifications and marketing (future)	Could

11.3 External Integrations (Future Roadmap)

Integration	Description	Target Phase
Payment Gateways	Card processing (Stripe, Mercado Pago, OpenPay)	Phase 2
Accounting Software	QuickBooks, Contpaq, Aspel	Phase 2
E-commerce	Shopify, WooCommerce, Mercado Libre	Phase 3
Banking	Automatic reconciliation	Phase 3
Delivery Platforms	Uber Eats, Rappi, Didi Food	Phase 3

12. Security & Compliance Requirements

12.1 Authentication & Authorization

1. Multi-factor authentication (MFA) support for owner and manager roles
2. Biometric authentication support on mobile devices (fingerprint, face recognition)
3. Passwordless login option via magic link (future)
4. Account lockout after 5 failed attempts; 15-minute cooldown
5. Password reset via verified email only

12.2 Data Protection

1. All customer personal data encrypted at rest
2. Right to data export (portability) per LFPDPPP
3. Right to data deletion (with audit log retention exception)
4. Data processing consent tracking for marketing communications
5. No sale of customer data to third parties (explicit policy)

12.3 Regulatory Compliance

1. **Mexico — SAT/CFDI:** System must generate CFDI 4.0 compliant invoices with correct XML structure, digital stamp, and QR code
2. **Mexico — IVA:** Automatic IVA calculation at 16% (standard) with support for 0% and exempt categories
3. **Mexico — LFPDPPP:** Privacy notice, consent management, data breach notification within 72 hours
4. **Pharmacy — COFEPRIS:** Controlled substance tracking with daily reporting capability
5. **Future — Brazil LGPD:** Architecture must support consent management and data subject rights

13. User Experience Requirements

13.1 Design System Principles

1. **Mobile-First:** Core workflows functional on 7" tablet; optimized on 10" tablet
2. **Touch-Optimized:** Minimum 44px touch targets; swipe gestures where intuitive
3. **Speed:** No loading spinners for primary actions; skeleton screens for secondary
4. **Feedback:** Every user action receives immediate visual confirmation
5. **Error Prevention:** Destructive actions require confirmation; forms validate inline
6. **Consistency:** Same patterns, colors, and behaviors across all modules

13.2 Accessibility Requirements

1. WCAG 2.1 Level AA compliance
2. Screen reader compatibility (ARIA labels on all interactive elements)
3. Keyboard-only navigation path for every feature
4. High contrast mode for visually impaired users
5. Font size adjustment (100%, 125%, 150%)
6. Color not used as sole information carrier (icons + text + color)

13.3 Localization Requirements

1. **Primary Language:** Spanish (Mexico) — all UI, error messages, reports, AI responses
2. **Secondary Language:** English — complete UI translation, reports in English
3. **Date Format:** DD/MM/YYYY (Mexico standard)
4. **Time Format:** 24-hour format
5. **Number Format:** 1,234.56 (comma thousands, point decimal)
6. **Currency:** MXN \$ (Mexican Peso) primary; USD \$ secondary for international
7. **Time Zone:** America/Mexico_City default; configurable per store

14. Business Rules

14.1 Inventory Rules

1. **BR-INV-001:** A product must have a unique SKU within a business.
2. **BR-INV-002:** A product barcode must be unique within a business unless explicitly configured as a variant.
3. **BR-INV-003:** Cost price cannot exceed sale price unless explicitly marked as “loss leader” with manager approval.
4. **BR-INV-004:** Inventory adjustments require a reason code; “Other” requires text explanation.
5. **BR-INV-005:** Negative stock is blocked by default; can be enabled per product with audit logging.
6. **BR-INV-006:** Expired products cannot be sold through POS; system blocks with warning.
7. **BR-INV-007:** Transfer between warehouses requires source quantity availability at time of execution.
8. **BR-INV-008:** Physical stock counts must be approved by a manager if variance exceeds 5% or \$1,000 MXN.

14.2 Sales Rules

1. **BR-SAL-001:** Every sale must have at least one item.
2. **BR-SAL-002:** Sale total must equal sum of item prices minus discounts plus taxes.
3. **BR-SAL-003:** Cashier cannot void their own transaction; requires supervisor PIN.
4. **BR-SAL-004:** Returns must reference original transaction; return quantity cannot exceed original quantity.
5. **BR-SAL-005:** Discounts exceeding 20% require manager approval.
6. **BR-SAL-006:** Fiado sales blocked if customer credit limit would be exceeded.
7. **BR-SAL-007:** Mixed payment must sum exactly to sale total; no partial payments unless Fiado.
8. **BR-SAL-008:** End-of-shift cash variance exceeding \$200 MXN requires manager investigation.

14.3 Purchasing Rules

1. **BR-PUR-001:** Purchase order must have at least one item and one supplier.
2. **BR-PUR-002:** Goods receipt quantity cannot exceed PO quantity by more than 10% without manager approval.
3. **BR-PUR-003:** Received cost price update triggers inventory valuation recalculation.
4. **BR-PUR-004:** Supplier invoice must match received goods within variance tolerance before payment approval.

14.4 Credit (Fiado) Rules

1. **BR-FIA-001:** New credit customers require owner/manager approval and signed agreement record.
2. **BR-FIA-002:** Credit limit default is \$1,000 MXN; maximum \$50,000 MXN without additional documentation.
3. **BR-FIA-003:** Accounts overdue 90+ days automatically flagged; new sales blocked unless payment made.
4. **BR-FIA-004:** Interest on overdue accounts calculated at 2% monthly (configurable) after grace period.
5. **BR-FIA-005:** Customer statements generated monthly and delivered automatically.

14.5 Security Rules

1. **BR-SEC-001:** Owner role cannot be deleted; only transferred.
2. **BR-SEC-002:** Audit logs cannot be modified or deleted by any user, including owners.
3. **BR-SEC-003:** Password must be changed every 90 days for admin roles (configurable).
4. **BR-SEC-004:** Inactive user accounts (90 days) automatically disabled.
5. **BR-SEC-005:** All financial data exports require owner or accountant role.

15. Acceptance Criteria

15.1 General Acceptance Criteria

1. **AC-GEN-001:** Every functional requirement must have a corresponding test case.
2. **AC-GEN-002:** Every test case must pass in both online and offline modes.
3. **AC-GEN-003:** No critical or high-severity bugs may exist at phase gate.

4. **AC-GEN-004:** Performance benchmarks must be met on target hardware (minimum: 4GB RAM Android tablet, Chrome browser).
5. **AC-GEN-005:** All user-facing text must be in Spanish (Mexico) with no untranslated strings.
6. **AC-GEN-006:** No placeholder text, dummy data, or “TODO” comments in production code.
7. **AC-GEN-007:** Security audit (OWASP ZAP or equivalent) must show zero critical vulnerabilities.
8. **AC-GEN-008:** Data migration from CSV/Excel must complete with < 1% error rate.

15.2 Module-Specific Acceptance Criteria

Inventory Module:

- AC-INV-001: 10,000 products imported; search returns correct result in < 500ms
- AC-INV-002: Barcode scan adds product to cart in < 1 second
- AC-INV-003: Inventory adjustment with reason; audit log shows before/after values

POS Module:

- AC-POS-001: Cashier completes 50-item sale in < 3 minutes
- AC-POS-002: Offline sale queues; syncs correctly upon reconnection
- AC-POS-003: End-of-shift report balances to the centavo

Fiado Module:

- AC-FIA-001: Credit sale blocked at exactly credit limit + \$0.01
- AC-FIA-002: Customer statement shows all transactions chronologically

Sync Module:

- AC-SYNC-001: 1,000 offline transactions sync without loss or duplication
- AC-SYNC-002: Conflict resolution UI presents local vs. server values clearly

16. Project Scope

16.1 In Scope (Phase 1 — MVP)

The following capabilities are explicitly in scope for the initial product release:

1. **Core Inventory Management:** Product catalog, stock tracking, adjustments, barcode support
2. **Point of Sale:** Touch-optimized POS, cash/card/Fiado payments, receipts, shifts
3. **Basic Purchasing:** Purchase orders, goods receipt, supplier linking
4. **Customer & Supplier Management:** Master data, history tracking
5. **Fiado (Credit):** Credit limits, balance tracking, payments, statements
6. **Basic Reporting:** Daily sales, inventory valuation, low-stock, fiado aging
7. **Role-Based Security:** Owner, Manager, Cashier, Warehouse roles
8. **Offline-First Operation:** Full offline functionality with sync
9. **Cloud Sync & Backup:** Automatic synchronization and daily backups
10. **Audit Logging:** Complete action tracking
11. **Settings & Configuration:** Business profile, tax, receipt customization
12. **AI Business Assistant:** Natural language queries, basic insights

13. **Multi-Store Support:** Up to 5 stores per business account
14. **Multi-Warehouse Support:** Up to 3 warehouses per store
15. **Dark/Light Mode:** Theme switching
16. **Spanish/English:** Complete UI translation

16.2 In Scope (Phase 2 — Growth)

1. Advanced analytics and dashboards
2. Mobile applications (iOS/Android native)
3. Electronic invoicing (CFDI) integration
4. Payment gateway integration (Mercado Pago, OpenPay)
5. Advanced promotions engine
6. Customer loyalty program
7. Employee scheduling and attendance
8. Advanced AI forecasting
9. Plugin/extension system (basic)
10. REST API for third-party integrations

16.3 In Scope (Phase 3 — Enterprise)

1. Unlimited stores and warehouses
 2. Supplier portal
 3. Customer portal
 4. Marketplace integration
 5. Advanced multi-currency
 6. Voice assistant
 7. Accounting software integrations
 8. Advanced regulatory compliance (LGPD, GDPR)
-

17. Out of Scope

The following capabilities are explicitly **not** in scope for Phase 1 and are deferred to future phases or excluded entirely:

1. **E-commerce Website Builder:** PAGATEC does not build online stores. Integration with existing e-commerce platforms is future scope.
2. **Payroll Processing:** Employee management includes basic records and performance, but payroll calculation, tax withholding, and payslip generation are out of scope.
3. **Advanced Accounting:** General ledger, double-entry bookkeeping, financial statement generation (balance sheet, cash flow) are out of scope. Basic P&L reporting is in scope.
4. **Hardware Manufacturing:** PAGATEC does not manufacture or sell barcode scanners, receipt printers, or POS hardware. Hardware compatibility lists are provided, but hardware sales are out of scope.
5. **Social Media Management:** No integration with Facebook, Instagram, or Twitter for marketing.
6. **Delivery Logistics:** No route optimization, driver tracking, or delivery fleet management.
7. **Manufacturing/BOM:** Recipe tracking for restaurants is in scope, but full manufacturing resource planning (MRP) with work orders and production scheduling is out of scope.

8. **Franchise Management:** Multi-store support is in scope, but franchise-specific features (royalty calculation, franchisee billing) are out of scope.
 9. **Cryptocurrency Payments:** Not supported in any phase currently planned.
 10. **Legal/Contract Management:** Supplier contracts and customer agreements are not managed within the system.
-

18. Release Roadmap

18.1 Phase 1: Foundation (Months 1-6)

Goal: Launch MVP for Mexican SMB market

Deliverables:

- Complete web application (responsive)
- Offline-first architecture
- Core inventory, POS, purchasing, fiado
- Basic reports and dashboard
- AI assistant (basic queries)
- Cloud sync and backup
- Role-based security
- Multi-store (up to 5)
- Spanish/English support

Success Gate: 100 beta businesses actively using; NPS ≥ 40 ; zero data loss incidents

18.2 Phase 2: Growth (Months 7-12)

Goal: Scale features and market presence

Deliverables:

- Mobile apps (iOS/Android)
- CFDI electronic invoicing
- Payment gateway integration
- Advanced analytics
- Loyalty program
- Employee scheduling
- Plugin system v1
- REST API v1
- Expansion to Colombia

Success Gate: 500 active businesses; 20% month-over-month growth; $< 2\%$ churn

18.3 Phase 3: Scale (Months 13-24)

Goal: Enterprise readiness and LATAM expansion

Deliverables:

- Unlimited stores/warehouses
- Supplier and customer portals
- Advanced AI forecasting
- Marketplace integration
- Accounting integrations
- Voice assistant
- Expansion to Chile, Peru, Argentina, Brazil
- LGPD/GDPR compliance

Success Gate: 5,000 active businesses; presence in 5 countries; profitability

19. Risk Analysis

19.1 Technical Risks

Risk ID	Risk	Probability	Impact	Mitigation
R-T-001	Offline sync conflicts corrupt inventory data	Medium	Critical	Robust conflict detection; manual resolution for complex cases; extensive testing with simulated network partitions
R-T-002	AI assistant hallucinates financial advice	Medium	High	RAG architecture with strict data grounding; confidence scores; human review for high-stakes recommendations
R-T-003	Performance degrades with large product catalogs (>50k)	Medium	High	Database indexing strategy; pagination; lazy loading; performance testing with synthetic large datasets
R-T-004	Cloud service downtime prevents sync	Low	Medium	Graceful degradation; extended offline capability; redundant infrastructure

Table 38 – continued

Risk ID	Risk	Probability	Impact	Mitigation
R-T-005	Security breach exposes customer business data	Low	Critical	OWASP compliance; penetration testing; encryption; least-privilege access; regular security audits
R-T-006	Mobile app battery drain due to sync	Medium	Medium	Optimized sync intervals; batch processing; background task management

19.2 Business Risks

Risk ID	Risk	Probability	Impact	Mitigation
R-B-001	SAT changes CFDI re-requirements, breaking invoicing	Medium	High	Modular CFDI integration; monitoring of regulatory changes; rapid update capability
R-B-002	Competitor (Square, Clip) launches similar LATAM-focused product	High	Medium	Differentiation through offline-first and AI; strong local support; pricing advantage
R-B-003	Target customers lack internet/devices to use cloud features	Medium	Medium	Offline-first design; low hardware requirements; tablet bundling partnerships
R-B-004	Customer data privacy concerns (LFPDPPP)	Medium	High	Transparent privacy policy; data encryption; local data storage option; consent management

Table 39 – continued

Risk ID	Risk	Probability	Impact	Mitigation
R-B-005	Key personnel (founder, lead developer) unavailable	Low	Critical	Documentation; knowledge transfer; bus factor mitigation; phased development reduces single-point-of-failure
R-B-006	Pricing model fails to achieve profitability	Medium	High	Multiple pricing tiers; usage-based options; enterprise upsell path; cost monitoring

19.3 Market Risks

Risk ID	Risk	Probability	Impact	Mitigation
R-M-001	Economic downturn reduces SMB software spending	Medium	High	Freemium tier; essential-only pricing; value demonstration (ROI calculator)
R-M-002	Latin American currency volatility affects pricing	High	Medium	Local currency pricing; annual plans with exchange rate locks; USD backup pricing
R-M-003	Low digital literacy in target segment slows adoption	Medium	Medium	Extreme simplicity; 15-minute onboarding; video tutorials; local partner training
R-M-004	Piracy/cracked software competition	High	Low	Cloud-dependent features; regular updates; value-added services (AI, support)

20. Assumptions & Constraints

20.1 Assumptions

1. **A-001:** Target businesses have at least one Android tablet or Windows PC with Chrome browser.

2. **A-002:** Internet connectivity, while intermittent, is available at least once per day for sync.
3. **A-003:** Business owners are motivated to reduce inventory losses and improve profitability.
4. **A-004:** SAT/CFDI requirements will remain stable enough to build against during Phase 1.
5. **A-005:** Barcode usage is prevalent or growing among target businesses.
6. **A-006:** Customers are willing to pay a monthly subscription for business software (SaaS model viable).
7. **A-007:** AI model APIs (OpenAI, Anthropic, or local) will remain available and cost-effective.
8. **A-008:** Target users can read and write in Spanish at a basic business level.
9. **A-009:** Cloud infrastructure costs (AWS, Azure, GCP, or local) will scale predictably.
10. **A-010:** Regulatory compliance requirements can be met without architectural redesign.

20.2 Constraints

1. **C-001:** Development budget and timeline are fixed per phase gates.
2. **C-002:** Target hardware minimum: 4GB RAM, 32GB storage, Android 8+ or Windows 10+ with Chrome.
3. **C-003:** Must support intermittent connectivity; cannot assume always-online operation.
4. **C-004:** Must comply with Mexican tax and data privacy laws before launch.
5. **C-005:** AI features must operate within reasonable token/cost limits per query.
6. **C-006:** Must be deployable on standard cloud infrastructure without custom hardware.
7. **C-007:** Must support both touch and mouse/keyboard interaction paradigms.
8. **C-008:** Must function on low-bandwidth connections (minimum 1 Mbps for sync).
9. **C-009:** Cannot require customers to purchase proprietary hardware.
10. **C-010:** Must achieve break-even within 24 months of launch.

21. Terminology & Business Glossary

21.1 Technical Terms

Term	Definition
API	Application Programming Interface; allows external systems to communicate with PAGATEC
CFDI	Comprobante Fiscal Digital por Internet; Mexico's electronic invoicing standard
CRUD	Create, Read, Update, Delete; basic data operations
ERD	Entity Relationship Diagram; visual representation of database structure
FEN	Forsyth-Edwards Notation; standard notation for chess positions (not applicable to PAGATEC, retained for completeness)
FIFO	First In, First Out; inventory valuation method
IVA	Impuesto al Valor Agregado; Mexico's value-added tax (16% standard rate)
JWT	JSON Web Token; secure method for transmitting authentication information
LIFO	Last In, First Out; inventory valuation method

Table 41 – continued

Term	Definition
LFPDPPP	Ley Federal de Proteccion de Datos Personales en Posesion de los Particulares; Mexico' s data privacy law
LGPD	Lei Geral de Protecao de Dados; Brazil' s general data protection law
MFA	Multi-Factor Authentication; security requiring multiple verification methods
MVP	Minimum Viable Product; the smallest version of the product that delivers value
NPS	Net Promoter Score; customer loyalty metric (-100 to +100)
OWASP	Open Web Application Security Project; standard for web application security
PGN	Portable Game Notation; standard for chess games (not applicable, retained)
POS	Point of Sale; the interface where sales transactions are processed
RAG	Retrieval-Augmented Generation; AI architecture that grounds responses in actual data
RBAC	Role-Based Access Control; permission system based on user roles
RFC	Registro Federal de Contribuyentes; Mexican tax identification number
SaaS	Software as a Service; cloud-hosted software accessed via subscription
SAT	Servicio de Administracion Tributaria; Mexico' s tax authority
SKU	Stock Keeping Unit; unique identifier for each product
SRS	Software Requirements Specification; formal document describing software requirements
TLS	Transport Layer Security; encryption protocol for data in transit
UI/UX	User Interface / User Experience; design and interaction quality
WCAG	Web Content Accessibility Guidelines; standards for accessible web content

21.2 Business Terms

Term	Definition
Abarrotes	Small grocery store; typical Mexican neighborhood shop
Aging Report	Report showing how long customer credit balances have been outstanding
Bill of Materials (BOM)	List of raw materials/components needed to produce a finished product
Centavo	Mexican currency subunit; 1 MXN = 100 centavos
COFEPRIS	Comision Federal para la Proteccion contra Riesgos Sanitarios; Mexican health regulatory agency
COGS	Cost of Goods Sold; direct costs attributable to production/sale of products
Credit Limit	Maximum amount a customer can owe on fiado before new sales are blocked

Table 42 – continued

Term	Definition
Fiado	Customer credit; selling goods with payment deferred to a future date
Inventory Shrinkage	Loss of inventory due to theft, damage, expiration, or administrative error
Loss Leader	Product sold below cost to attract customers who will buy other profitable items
Mini-Market	Small retail store larger than a convenience store but smaller than a supermarket
Overstocking	Holding more inventory than needed, tying up capital and increasing waste risk
Purchase Order (PO)	Commercial document issued to a supplier indicating types, quantities, and agreed prices
Reorder Point	Inventory level at which a new order should be placed
Stock-Out	Situation where inventory for a product is depleted and unavailable for sale
Tiendita	Very small convenience store; often family-operated
Variance	Difference between expected and actual values (stock count, cash, delivery)

22. Measurable Success Criteria

22.1 Product Success Metrics

Metric	Target	Measurement Method
Time to First Sale (new user)	< 15 minutes	Onboarding analytics
Feature Adoption Rate	> 70% of features used within 30 days	Usage analytics
User Retention (30-day)	> 80%	Cohort analysis
User Retention (90-day)	> 60%	Cohort analysis
Net Promoter Score (NPS)	>= 50	Quarterly survey
Customer Satisfaction (CSAT)	>= 4.5/5	Post-interaction survey
Support Ticket Volume	< 1 per user per month	Helpdesk metrics
Time to Resolve Support Issue	< 24 hours	Helpdesk metrics

22.2 Business Success Metrics

Metric	Phase 1 Target	Phase 2 Target	Phase 3 Target
Active Businesses	100	500	5,000
Monthly Recurring Revenue (MRR)	\$10,000 USD	\$50,000 USD	\$500,000 USD
Churn Rate	< 5%/month	< 3%/month	< 2%/month
Average Revenue Per User (ARPU)	\$100 USD/month	\$100 USD/month	\$100 USD/month
Customer Acquisition Cost (CAC)	< \$50 USD	< \$40 USD	< \$30 USD
Lifetime Value (LTV)	> \$1,000 USD	> \$1,500 USD	> \$2,000 USD
LTV:CAC Ratio	> 3:1	> 5:1	> 10:1

22.3 Technical Success Metrics

Metric	Target	Measurement Method
Application Load Time	< 3 seconds	Lighthouse CI
API Response Time (p95)	< 200ms	APM monitoring
Uptime	99.9%	Uptime monitoring
Sync Success Rate	99.99%	Automated sync testing
Data Loss Incidents	0	Incident tracking
Security Vulnerabilities (Critical)	0	Penetration testing
Test Coverage	> 80%	CI/CD pipeline
Deployment Frequency	Daily	CI/CD pipeline
Mean Time to Recovery (MTTR)	< 4 hours	Incident tracking

23. Missing Requirements & Gaps Identified

During the analysis of the initial project brief, the following gaps were identified and must be addressed in subsequent phases or through executive decision:

23.1 Critical Gaps (Must Resolve Before Phase 1)

- 1. **GAP-001 — Pricing & Subscription Model:** No pricing tiers, freemium boundaries, or payment collection mechanism are defined. Decision required: Will there be a free tier? What features are paid? What is the monthly price per store/user?

2. **GAP-002 — Hardware Compatibility Matrix:** No specification of supported barcode scanners, receipt printers, cash drawers, or label printers. Decision required: Which hardware brands/models are certified? USB, Bluetooth, or network printers?
3. **GAP-003 — Data Migration Strategy:** No plan for migrating customers from existing systems (Excel, old POS, notebooks). Decision required: Will migration service be offered? Self-service import only? Data mapping templates?
4. **GAP-004 — Customer Support Model:** No definition of support channels, hours, SLAs, or escalation paths. Decision required: WhatsApp support? Phone? Email? Business hours only or 24/7?
5. **GAP-005 — Disaster Recovery Plan:** While backups are mentioned, no formal DR plan exists. Decision required: RPO (Recovery Point Objective) and RTO (Recovery Time Objective) definitions? Geographic redundancy?

23.2 High-Priority Gaps (Resolve During Phase 1)

6. **GAP-006 — Training & Onboarding Program:** No curriculum, video tutorials, or knowledge base structure defined. Recommendation: Produce 10-video onboarding series; in-app guided tours.
7. **GAP-007 — Multi-Currency Support:** Brief mentions MXN and USD but no mechanism for dual-currency transactions or exchange rate handling. Recommendation: Define if multi-currency is Phase 1 or Phase 2.
8. **GAP-008 — Expense Tracking:** “Financial Dashboard” is listed but no explicit expense recording (rent, utilities, salaries) outside inventory COGS. Recommendation: Add expense categories and manual expense entry.
9. **GAP-009 — Supplier Payment Tracking:** Purchasing module tracks POs but not actual payments to suppliers. Recommendation: Add accounts payable tracking.
10. **GAP-010 — Customer Communication:** No SMS, email, or WhatsApp integration for marketing or payment reminders. Recommendation: Phase 2 integration with Twilio or similar.

23.3 Medium-Priority Gaps (Resolve During Phase 2)

11. **GAP-011 — Advanced Promotions:** “Promotions” listed as “Could” priority but no detailed requirements (time-based, customer-segment, BOGO). Recommendation: Define promotion engine requirements before Phase 2.
12. **GAP-012 — Recipe/BOM for Restaurants:** Mentioned in Luis persona but not in core modules. Recommendation: Add recipe management to Phase 1 if restaurant segment is priority.
13. **GAP-013 — Appointment/Service Booking:** Service businesses listed as target but no service scheduling module. Recommendation: Phase 2 or Phase 3.
14. **GAP-014 — Advanced Analytics:** Current analytics are basic. No predictive models, cohort analysis, or cohort retention metrics defined. Recommendation: Define AI model requirements and data science pipeline.
15. **GAP-015 — API Rate Limits & Documentation:** REST API mentioned for Phase 2 but no rate limits, authentication methods, or versioning strategy defined. Recommendation: Define API governance before Phase 2.

23.4 Low-Priority Gaps (Future Consideration)

- 16. **GAP-016 — Franchise Management:** Multi-store support exists but franchise-specific features (royalties, franchisee agreements) are undefined.
- 17. **GAP-017 — Manufacturing/MRP:** Recipe tracking exists for restaurants but manufacturing with work orders, BOM explosion, and capacity planning is out of scope. Clarify if this ever enters scope.
- 18. **GAP-018 — White-Labeling:** No mention of white-label options for distributors or chains wanting branded software. Consider for Phase 3.
- 19. **GAP-019 — Offline Data Retention Limits:** No specification of how much data is stored locally vs. purged. Recommendation: Define local storage limits and auto-archive policy.
- 20. **GAP-020 — Accessibility Certification:** WCAG compliance mentioned but no formal accessibility audit or certification plan. Recommendation: Budget for third-party accessibility audit in Phase 2.

Document Approval

This Product Vision & Business Requirements Document has been reviewed and approved by the project stakeholders. It serves as the constitutional foundation for all PAGATEC development activities.

Role	Name	Signature	Date
CEO / Product Owner	Yeney B. Herrera	_____	2026-07-30
Engineering Lead	TECCEF.CORP Engineering Team	_____	2026-07-30

End of Document Total Sections: 23 Document Classification: Internal — Project Constitution Next Action: Phase 1 Gate Review upon completion of Software Requirements Specification (SRS) detailed elaboration.